INTERNAL CONCEPT PROPOSAL · SIEMENS MANUFACTURING USE CASE

GiNi Production Flow & Recovery

Seeing how one disruption affects the whole shift



People
who is present and qualified

Tanks
each with its own ID

Work centers
WC1 → WC4 (provisional)

Inspections
pass · fail · incomplete

Decisions
feasible recovery actions

October 2026

The question Rajesh raised

One upstream gap can reach every later station but not all at once

If one work center is disrupted, what happens to the rest of production, when does the impact arrive and what can the supervisor do?

Our paraphrase of the discussion with Rajesh — not a verbatim quote

WC1 Pre-Assembly — Crew 4 of 5 — IMPACT: NOW

WC2 Assembly — Crew complete — IMPACT: LATER?

WC3 Painting — Crew complete — IMPACT: LATER?

WC4 Markings — Crew complete — IMPACT: LATER?

Capacity

How much slower is WC1 with the people and skills it actually has?

Queues

How many approved tanks are already waiting downstream?

Timing

When will each later station actually run out of work?

A normal tank journey

Provisional route for the demo to be confirmed with Siemens

TANK-0014

WORKING ASSUMPTION

Production order
ORD-1042

→

WC1
Pre-Assembly

Gate 1 · first-stage fail = permanent scrap

→

WC2
Assembly

Checks here failure rule TBC

→

WC3
Painting + cure

Checks here failure rule TBC

→

WC4
Markings

Checks here failure rule TBC

→

Tank route
complete

Fail at gate 1 → tank scrapped permanently (confirmed rule)

Later-stage fail → hold until an approved disposition (rule to confirm)

Paint cure is process time — never counted as worker idle time

Reference structure — outside the first demo

WC5–WC6
Cover stream

WC7–WC8
Accessories stream

Where and how these join the tank is not confirmed — no join drawn

"Tank route complete" ≠ finished product ready to dispatch

What we know and what needs confirmation

Separate what is decided, what the sources suggest, and what Siemens still needs to answer

Confirmed requirement

From our brief and the discussion

- First-stage fail = permanent scrap
- Scrapped tank keeps its history and is never reused
- Replacement = new physical tank, new ID, linked to the original
- Forecast updates as soon as the fail is recorded
- Incomplete inspection ≠ failed inspection

Source-supported structure

From our project notes on the CTQ workbook (workbook not attached to this review)

- Tank stream: WC1–WC4
- Cover stream: WC5–WC6
- Accessories stream: WC7–WC8
- CTQ review stages are tentative

Demo assumptions / open questions

Need Siemens confirmation

- Exact route and inspection gates
- What happens after a later-stage fail
- Times, crew sizes and skills (demo values only)
- How covers and accessories join the tank
- Queue limits, material rules, order priority

This is an internal proposal. Siemens has not reviewed, approved or endorsed it.

Meet the supervisor’s morning

06:00 — one shift, one order, a small cast

ALL CHARACTERS FICTIONAL

Maya
Shift supervisor
Makes the calls

Arun
Markings worker, WC4
Also qualified for WC1

Leena
Inspector, WC1 gate
Records inspection results

Ravi
WC1 worker
Absent today

TANK-0018
Tank on ORD-1042
Will fail first-stage inspection

TANK-0021
Possible replacement
Exists only if feasible

Shift plan at 06:00

SYNTHETIC SCENARIO

Shift
06:00–14:30 · break 10:00–10:30

Order ORD-1042
10 good tanks, tank route complete by 14:30

Already approved
3 tanks waiting at WC2 (TANK-0011–0013)

To start at WC1
7 tanks (TANK-0014–0020)

Planned crew
WC1 5 · WC2 4 · WC3 2 · WC4 2

Normal-day forecast
10th tank done 12:45 · 105 min to spare

Disruption one: a worker is unavailable

Ravi's absence slows WC1 — but the order is not late yet

Planned WC1 crew



Available today



Ravi — absent
confirmed 06:05

Configured demo rule (simplified)

- WC1 needs 200 worker-minutes per tank
- 5 people → 40 min per tank · 4 people → 50 min
- Minimum crew 3 — below that WC1 stops
- Only WC1-qualified people count

Not a general law: real operations may need a fixed crew or may not split. Each operation gets its own rule from Siemens standards.

WC1 finishes its 7 tanks

12:20

normal day: 11:10

10th good tank

13:55

normal: 12:45 · still on time

Margin to 14:30

35 min

normal day: 105 min

Assembly waiting (full crew)

85 min

normal day: 15 min

Maya

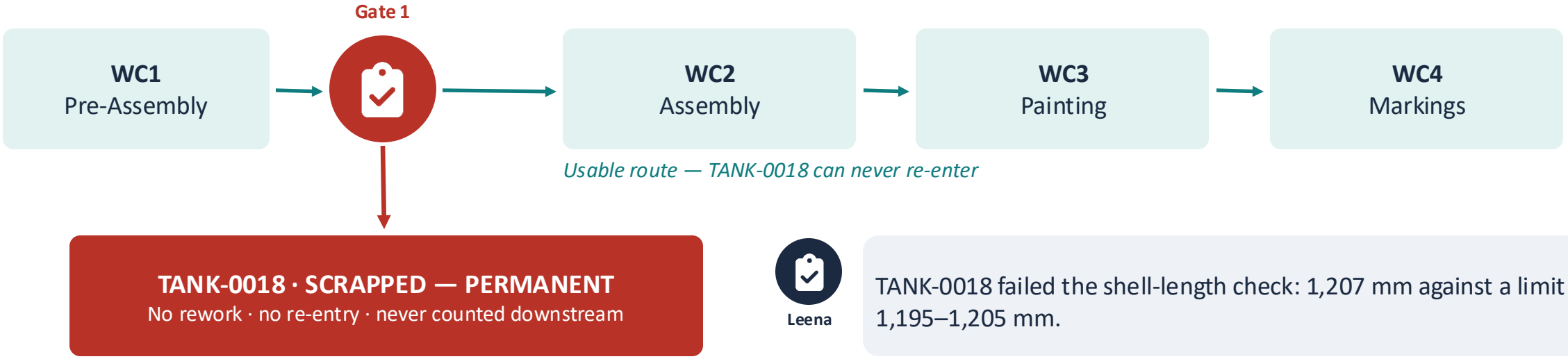
We're one short at WC1. Are we going to be late?

GiNi

Not yet. The order still finishes at 13:55 — but the margin drops from 105 to 35 minutes.

Disruption two: a tank fails inspection

TANK-0018 leaves the usable route permanently at 10:40



History kept

- Inspection readings and inspector (Leena)
- 200 worker-minutes of WC1 labour
- Material kit from lot ML-0412
- Linked to ORD-1042 and to any replacement

Order impact

- Failed 10:40 · recorded 10:48
- ORD-1042 can now reach at most 9 of 10 good tanks
- Forecast recalculated as soon as the result is recorded
- Root cause investigated separately — it doesn't hold up the forecast

Why downstream work does not stop immediately

WIP = tanks already waiting or being worked on. It buys time — until it runs out.



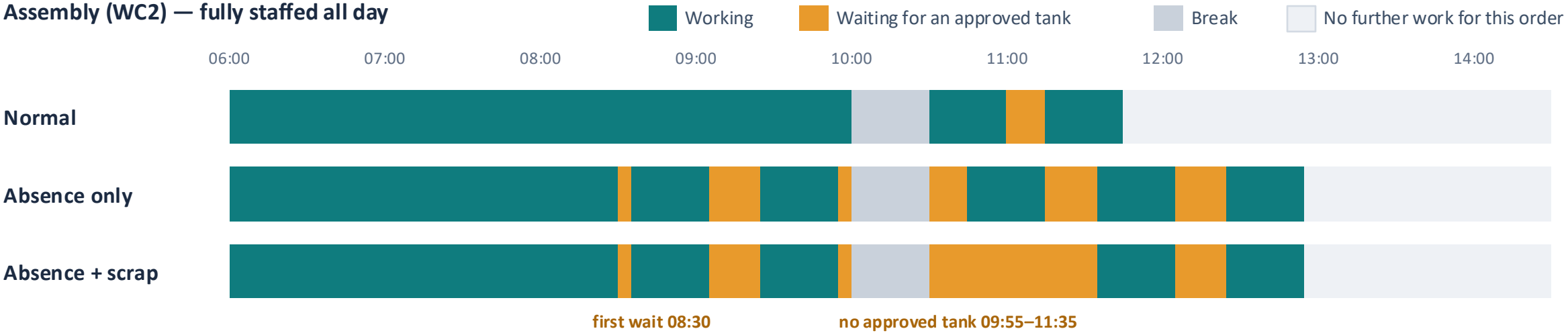
Maya

Assembly has its full team. Why is it running out of work?

GiNi

Its approved queue is being used faster than WC1 can refill it.

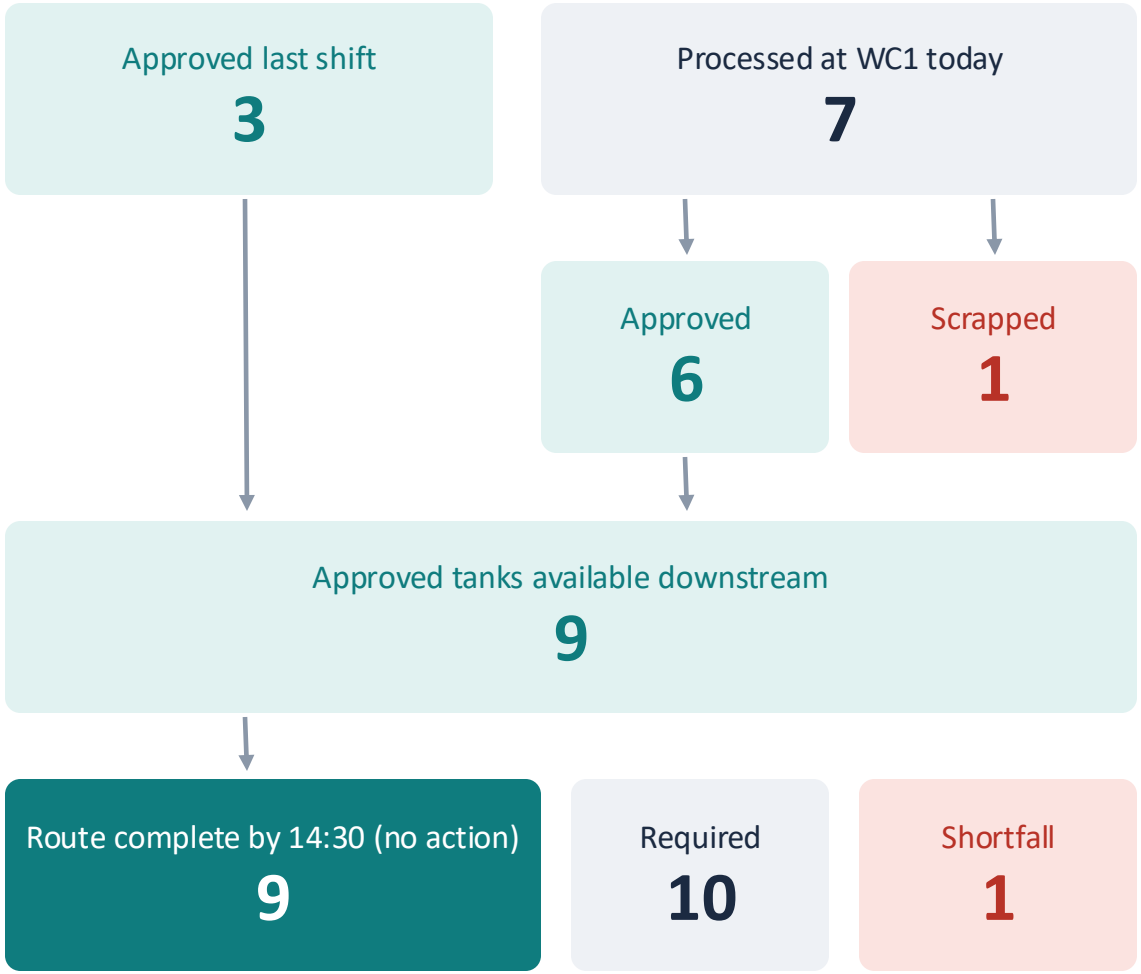
Assembly (WC2) — fully staffed all day



Waiting with a full crew of 4: Normal 15 min · Absence only 85 min · **Absence + scrap 115 min (≈ 7.7 worker-hours)**

What reaches the order deadline?

Count each tank once — at the end of the route



WC1 releases are not finished tanks.

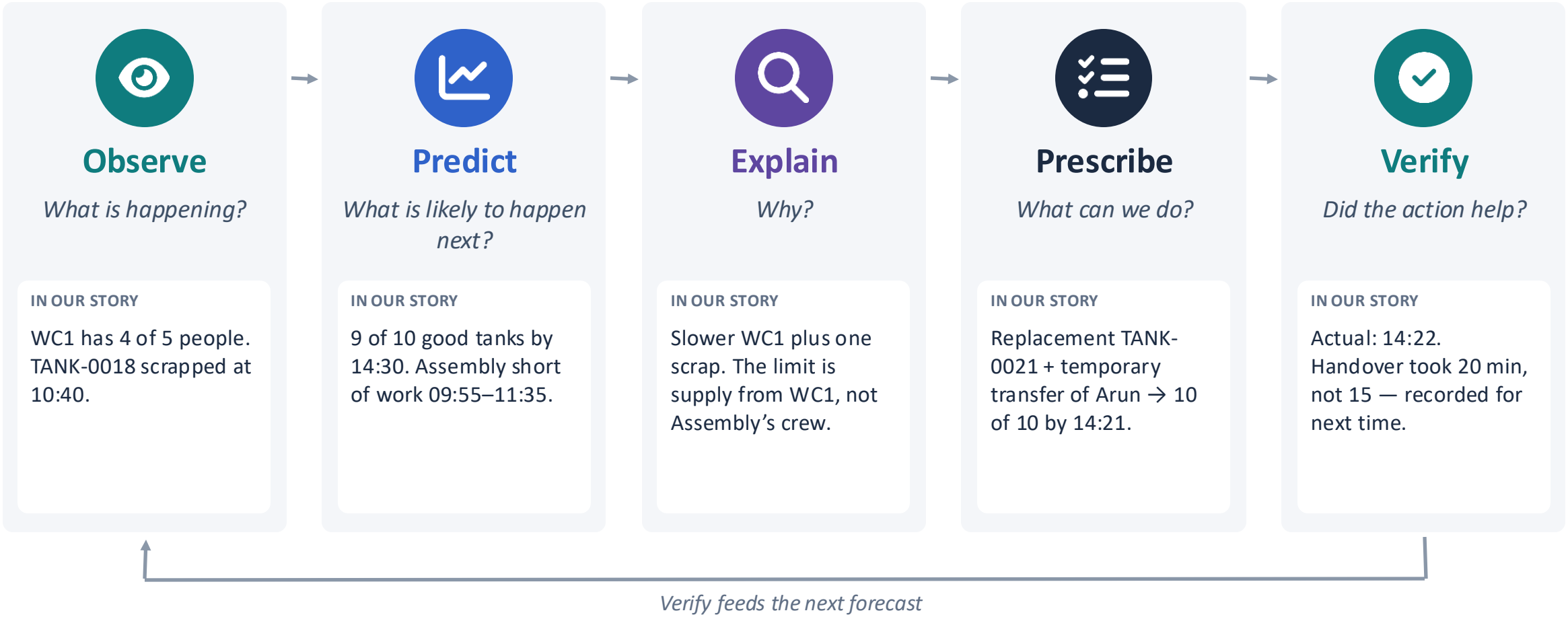
Separately vs together

Scenario	Good by 14:30	10th tank done
Normal	**10**	12:45
Absence only	**10**	13:55
Scrap only + replacement	**10**	13:25
Absence + scrap, no action	**9**	—
Absence + scrap + replacement	**9**	**14:45 (late)**

Each disruption alone is recoverable. Together they are not: the slower WC1 can only start the replacement at 12:20, so it misses 14:30 by 15 minutes. Effects interact — they don't simply add.

The five application capabilities

Everyday questions, answered in order — and checked afterwards



The supervisor’s main screen

Maya’s view at 10:55 — one place for the shift, the order and the alerts

CONCEPT MOCKUP · FICTIONAL DATA · NOT A LIVE SCREEN

Shift overview · Line A

Selected time 10:55 · Data as of 10:54 · Last event received 10:48

Target	Good completed	Forecast good by 14:30	Scrapped	Orders at risk
10 good tanks by 14:30	6 route complete	9 at risk	1 TANK-0018	1 ORD-1042

WC1 Pre-Assembly

SHORT-STAFFED

Crew 4 of 5
TANK-0019 in work, done ~11:30
1 tank waiting

WC2 Assembly

WAITING FOR WORK

Crew 4 of 4
No approved tank since 09:55
Next tank ~11:35

WC3 Painting

NORMAL

Crew 2 of 2
TANK-0017 curing until 11:10

WC4 Markings

NORMAL

Crew 2 of 2
Next tank ~11:15

ORD-1042 · 10 good tanks by 14:30

6 done	3 in progress	1 scrapped

Forecast: 9 of 10 by 14:30 — short by 1. Recovery options available.

Alerts · 3 open

10:48 · TANK-0018 failed first-stage check — scrapped
Owner: Maya · open

ORD-1042 forecast 9 of 10 by 14:30
Recovery options ready · open

WC2 out of approved tanks until ~11:35
Cause: WC1 supply · open

Acknowledging an alert does not close it.

Click a tank

A passport per tank: route, people, readings, outcome

CONCEPT MOCKUP · FICTIONAL DATA · NOT A LIVE SCREEN

Tank passport · TANK-0018

Selected time 10:55

ORDER
ORD-1042

ROUTE
Tank route v0.1 (provisional)

MATERIAL KIT
Lot ML-0412

LABOUR CONSUMED
200 worker-minutes

BLOCKERS
None recorded

STATUS
SCRAPPED — cannot proceed

REPLACED BY
TANK-0021

Prior shift	Registered · linked to ORD-1042 and kit lot ML-0412
09:20	WC1 Pre-Assembly started · crew: Sana, Dev, Meera, Kiran
10:00–10:30	Paused · shift break
10:40	WC1 inspection by Leena Shell length 1,207 mm (limit 1,195–1,205) — FAIL Weld visual — pass · Flange position — pass
10:40 · rec. 10:48	Disposition: SCRAP (first-stage rule)
10:55	Replacement TANK-0021 approved by Maya

Click a work center

Assembly is fully staffed — its problem is supply, not people

CONCEPT MOCKUP · FICTIONAL DATA · NOT A LIVE SCREEN

Work center · WC2 Assembly

Selected time 10:55 · Data as of 10:54

Crew today · 4 of 4 present

Priya	Present · Assembly qualified	WAITING · NO TANK
Tom	Present · Assembly qualified	WAITING · NO TANK
Lin	Present · Assembly qualified	WAITING · NO TANK
Ahmed	Present · Assembly qualified	WAITING · NO TANK

Equipment

Assembly lift AL-1	AVAILABLE

Skill needed: Assembly — all four qualified

Approved tanks waiting

0

since 09:55

Active tanks

0

Next arrival (forecast)

~11:35

TANK-0019 from WC1

LIMITING CONSTRAINT

Upstream supply from WC1

— not labour at WC2

Adding people here would not help.

Waiting reason recorded: no approved tank (upstream). Not a performance issue for this crew.

Click a worker

Observed and forecast kept separate, with a reason for every wait

CONCEPT MOCKUP · FICTIONAL DATA · NOT A LIVE SCREEN

Worker · Arun — Selected time 11:40 · Data as of 11:39

Arun

Home: WC4 Markings
Present since 06:00

QUALIFICATIONS

WC4 Markings · valid
WC1 Pre-Assembly · valid

Shown with the reason for any wait — never ranked by raw tank counts.

If scans stop, the screen shows “last known: …” — never “idle”.

OBSERVED NOW

Working at WC1 Pre-Assembly on TANK-0020 · shell fit-up · crew of 5 (joined 11:15)

Today, 06:00–11:40 (observed)



Active work	No work available	Break	Transfer + handover
85 min	205 min	30 min	20 min
WC4 60 · WC1 25	reason: no tanks at WC4	10:00–10:30	plan was 15

CREW CONTRIBUTIONS

6 markings operations shared in a crew of 2; WC1 work shared in a crew of 5 — not counted as individual tanks

FORECAST

Free ~12:47 when TANK-0021 leaves WC1; back at WC4 ~12:52

Where was the problem detected, and where did it originate?

Detection tells us where a defect was noticed — not who or what caused it

OPERATOR / CREW

WC1 fit-up crew: Sana, Dev, Meera, Kiran LINKED ONLY

EQUIPMENT

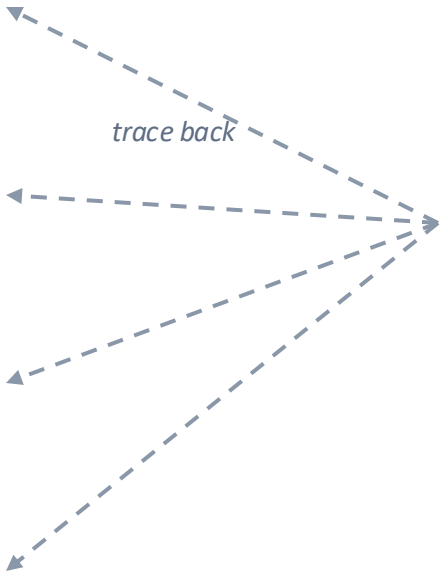
Fixture FX-02 LINKED ONLY

MATERIAL

Shell sheet lot ML-0412 LINKED ONLY

PROCESS

Work instruction WI-PA-07 v3 LINKED ONLY



DETECTED HERE

WC1 inspection gate

10:40 · by Leena (inspector)

Leena found the defect. She is the detector, not a suspect.

No origin confirmed yet.

Detected at ≠ caused by. **Performing an operation ≠ causing the defect.**
Detector, operator/crew, equipment, material and process are recorded as separate fields.

How responsibility becomes confirmed

A short, explicit process — and "unknown" is an acceptable answer

Report	Trace evidence	Investigate	Confirm cause	Assign corrective action
10:48	**11:10**	**12:10**	**13:20**	**13:30**
TANK-0018 fail recorded. Cause not yet known.	Linked to fit-up crew, fixture FX-02, lot ML-0412 and WI-PA-07. Fixture and fit-up step suspected.	Joseph (quality engineer) finds the FX-02 clamp offset; TANK-0017 reading also near the upper limit.	Equipment cause confirmed by Joseph, with evidence attached. Crew recorded: no fault.	Maintenance recalibrates FX-02 (owner: maintenance lead). Tracked until closed.
UNKNOWN	SUSPECTED	SUSPECTED	CONFIRMED	ACTION OWNED

Not enough evidence? Stay suspected or unknown, collect more, review again

A worker–operation link alone never establishes fault. Responsibility is shown only after a named reviewer confirms it with evidence.

Four different causes, four different responses

The right response depends on the confirmed cause — not on who was nearest



Confirmed worker-related task error

Evidence Investigation confirmed a missed or incorrect step, with a named reviewer.

Response Coaching or retraining; review the instruction. Visible only to authorised roles.



Equipment-caused defect

Evidence Fixture or machine fault confirmed — our TANK-0018 case (FX-02).

Response Maintenance and containment; check other tanks made on it. Crew not at fault.



Upstream shortage → downstream waiting

Evidence No approved tank in the queue; crew present and qualified.

Response Fix supply upstream: transfer, relief, replacement. Waiting is not underperformance.



Unresolved cause

Evidence Evidence is incomplete or conflicting.

Response Keep it “unknown”, collect evidence, monitor. No responsibility assigned.

Recovery options

At 10:55 — each option, the constraint it can address, and its result in this scenario

Qualified transfer

Addresses Labour short at WC1

Arun from WC4 — needed together with a replacement

PART OF BEST PLAN

Relief worker

Addresses Labour short at WC1

Free only from 12:30 → with replacement, 10th tank 14:37

7 MIN LATE

Replacement tank

Addresses A scrapped good unit

Alone → 10th tank 14:45; slow WC1 can't start it sooner

15 MIN LATE

Permitted resequencing

Addresses Which order goes first

No other order this shift

NOT APPLICABLE

Approved overtime

Addresses Time available

Extends the shift, not the 14:30 due time; needs approval

MISSES DUE TIME

No action

Addresses —

9 of 10 good tanks by 14:30

SHORT BY 1

Best feasible plan: replacement TANK-0021 + temporary transfer of Arun → forecast 10 of 10 by 14:21.

Adding people only helps when labour is the constraint.

Test the move before approving it

Every transfer is checked on both stations — the one that gains and the one that loses

Check	Arun · WC4 Markings → WC1	Priya · WC2 Assembly → WC1
Qualified for WC1 pre-assembly	**Yes** valid qualification	**Yes** valid qualification
Available	**Yes** from 10:55	**Yes** from 10:55
Travel + handover	15 min — joins WC1 at 11:10	10 min
Source keeps its minimum crew	**Yes** WC4 2 → 1 (minimum 1)	**No** WC2 4 → 3 (minimum 4): Assembly stops
Source-station impact	WC4 10 → 20 min per tank; TANK-0017 done 11:35, not 11:25; no order late	Assembly cannot run at all
Equipment and material at WC1	**Yes** fixture FX-01 and kits available	**Yes** same
Good tanks by 14:30 (with replacement)	10 of 10 · last tank 14:21	Worse — no assembly
Decision	**ACCEPT — temporary, back at WC4 ~12:51**	**REJECT**

A replacement is a new production commitment

Approved at 10:55 — route complete at 14:21. It does not appear instantly.

TANK-0018
Scrapped 10:40 · history kept

replaced by

TANK-0021
New ID · linked to TANK-0018 and ORD-1042

Maya
Can we replace it before the order deadline?

GiNi
Let's check material, qualified people and the remaining route time.

Feasibility checks

- Material: 1 spare kit (lot ML-0415)
- Approved by Maya at 10:55
- WC1 can only start it at 12:06, after TANK-0020
- Must pass the WC1 gate — the forecast assumes it does
- Route complete 14:21 — before 14:30



Transfers of 5 min between stations shown as gaps. If TANK-0021 fails its WC1 gate, the shortfall returns.

Before action, after action, actual outcome

Verify keeps the forecast honest — predicted and observed are recorded separately

Measure	Before action forecast at 10:55	After action forecast at 10:55	Actual synthetic replay
Good tanks by 14:30	9 of 10	**10 of 10**	**10 of 10**
10th good tank completes	not this shift	**14:21**	**14:22**
Margin to deadline	short by 1	9 min	8 min
Arun joins WC1	—	11:10	11:15
Handover time	—	15 min	20 min
WC4 time per tank while he's away	10 min	20 min	20 min
Assembly waiting	7.7 worker-h	7.4 worker-h	7.5 worker-h

"Actual" = the same scenario replayed with a 20-minute handover. Illustration only, not an observed factory result.

Predicted ≠ observed

The handover took 20 minutes, not 15, so the last tank finished a minute later. Recorded, so the next similar transfer is forecast better.

Not cash savings

Waiting worker-hours are capacity that could be redeployed — not proven savings. Here the action recovered the order, not the idle time.

Action states kept separate

Proposed 10:52 · approved 10:55 · started 10:55 · completed 11:15

What we can reuse

Existing GiNi patterns feed a new manufacturing engine

Live Shift

gini_live_shift

Worker state, shift timelines, operational events, replay

Variance demo

gini_smd_variance_demo_clear_ui

Variance detection, alerts, explanations, scenario validation

SMD

SMD-mainF

Worker eligibility, safe reallocation, before-and-after comparison

New: manufacturing flow engine

- Provisional route WC1–WC4 and dependencies
- Tank-level traceability
- Inspection gates and dispositions
- Permanent first-stage scrap
- Replacements with new IDs
- Event-based forecast and recovery comparison

REUSE CANDIDATES · NOT VERIFIED IN THIS REVIEW

Repositories were not attached; capabilities as described in our project notes. Patterns, not merged code — and not Siemens features.

What data makes this possible?

Six groups, one central chain, and two times on every event

People

roster, skills, attendance, tasks

Process

route, times, crew rules

Tanks & orders

tank IDs, orders, WIP

Quality

checks, readings, decisions

Materials & equipment

kits, lots, fixtures, downtime

Actions & evidence

transfers, approvals, investigations

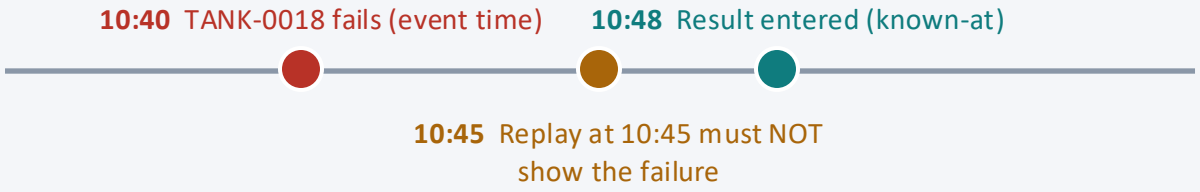
One relationship answers most questions

WORKER **Arun** → TASK **WC1 fit-up** → TANK **TANK-0020** → INSPECTION **WC1 gate** → ORDER **ORD-1042**

Two times on every event

Event time: when it happened
Known-at time: when the system learned it

A replay must never show what wasn't yet known.



The synthetic demo and its test cases

Reproducible, fictional, and checked against separately calculated answers

1	~20	20	1	15
tank route (WC1–WC4)	fictional workers + relief	days of illustrative history	scripted disruption shift	test scenarios

Normal shift	WC1 absence	First-stage scrap	Combined disruption	Protective WIP
Exhausted WIP	Successful transfer	Rejected transfer	Feasible replacement	Replacement material missing
Machine breakdown	Delayed data	Confirmed worker cause	Equipment cause	Unresolved cause

Flow · Recovery · Operations & data · Cause tracing

SIMULATION — not live factory operation. Every scenario gets an expected result calculated separately from the engine. For this deck, a second, independently written calculation reproduced all seven key scenario results.

Questions to close and phased delivery

Answers from Siemens decide the build; real data decides whether forecasts can be trusted

Questions to close with Siemens (priority order)

1. Actual tank route and inspection gates
2. What happens to a tank that fails after WC1
3. Standard times, crew and skill rules
4. How queues and WIP behave; queue limits
5. Material availability and replacement approval
6. Order priority and deadline rules
7. How covers and accessories join the tank

Phased delivery

1

Prototype on synthetic data

Proves the rules, calculations and supervisor experience for WC1–WC4.

2

Shadow pilot on real data

Runs alongside real shifts; forecasts compared with what actually happens. No decisions driven by it.

3

Production deployment

Integration with factory systems, roles and governance — only after the pilot.

Synthetic validation proves the rules are applied correctly — not real-world forecast accuracy.

The decision we need

Six internal agreements to move forward

- **WC1–WC4 demo scope**
  Tank route only; covers and accessories later.
- **Permanent first-stage scrap**
  A fixed rule in the engine and the data.
- **Worker and task traceability**
  With the fairness rules: no blame without confirmed evidence.
- **Assumptions and process owners**
  A register with a named owner for each open question.
- **A sample real tank history**
  One real shift to request from Siemens.
- **Demo acceptance criteria**
  Agreed before we build.

From “something is wrong at WC1” to “here is what it means for the order, when it hits, why, and which action is feasible” — in one place, and checked afterwards.

A1 · Scenario assumptions behind every number

Assumption	Demo value
Shift	06:00–14:30; one break 10:00–10:30
During the break	Labour pauses; transfers and paint cure continue
Order ORD-1042	10 good tanks, tank route complete by 14:30
Opening WIP	TANK-0011–0013 approved, waiting at WC2
WC1 Pre-Assembly	200 worker-min/tank: 5 people 40 min, 4 people 50 min; minimum 3
WC2 Assembly	30 min/tank with 4; minimum crew 4
WC3 Painting	20 min/tank with 2, then 20 min cure
WC4 Markings	10 min/tank with 2, 20 min with 1; minimum 1
Between stations	5 min transfer; first in, first out; one tank at a time per station
Inspection record	Entered 8 min after the decision
Arun's transfer	Handover 15 min (plan) / 20 (actual); back 5 min after TANK-0021 leaves WC1
Relief / replacement	Relief free from 12:30 · 1 spare kit · replacement assumed to pass WC1

Scenario	Good by 14:30	Last tank	WC2 wait (min)
Normal	**10**	12:45	15
Absence only	**10**	13:55	85
Scrap only, no replacement	**9**	12:45	45
Scrap only + replacement	**10**	13:25	55
Absence + scrap, no action	**9**	13:55	115
…+ replacement only	**9**	14:45	135
…+ replacement + relief	**9**	14:37	127
…+ replacement + Arun (plan)	**10**	14:21	111
…+ replacement + Arun (actual)	**10**	14:22	112

Model: minute-step flow model. Cross-check: a separately written interval calculation reproduced every key result above. Both scripts are delivered alongside the deck.

A2 · Synthetic data (1 of 3): process, capacity and people

A · Process and capacity

Table	Essential columns
PRODUCT_MASTER	product ID, model, component type, route and version
WORK_CENTER_MASTER	center ID and name, queue capacity
ROUTING_OPERATIONS	operation ID, sequence, work center, elapsed and labour minutes, setup, minimum crew, skill, equipment, quality gate
ROUTE_DEPENDENCIES	predecessor, successor, transfer time, release condition
CREW_CAPACITY_RULES	operation/product, crew count, duration, parallel capacity
EQUIPMENT_MASTER + calendars	equipment ID, supported operations, capacity, availability, breaks, downtime

B · People

Table	Essential columns
WORKER_MASTER	ID, fictional name, role, home station
WORKER_QUALIFICATIONS	skill, valid from/to, status
SHIFT_ROSTER	worker, shift, planned assignment and times
WORKER_EVENTS	attendance, work, break, support, transfer, confirmed absence
TASK_ASSIGNMENTS	tank, operation/subtask, worker, assigned and accepted time
OPERATION_CREW + intervals	worker, role, joined/left, pauses and reasons
HANDOVERS	from/to operation, people, time, condition, evidence

A3 · Synthetic data (2 of 3): production, materials and quality

C · Production and materials

Table	Essential columns
PRODUCTION_ORDERS	order/line, product, required good qty, release, due time, priority
UNIT_MASTER	tank ID, order, route, material lot, replacement-for ID
OPERATION_INSTANCES	tank, operation, attempt, equipment, batch
PRODUCTION_EVENTS	start, pause, resume, finish, transfer, route completion
OPENING_WIP	tank, stage, remaining work, previous releases
MATERIAL_REQUIREMENTS / LOTS / EVENTS	needs per product, lots on hand, issues and returns
REPLACEMENT_REQUESTS	original unit, approval, new unit, timing

D · Quality and cause tracing

Table	Essential columns
CTQ_CHECK_MASTER	check/version, source row, question, failure answer, units, nominal/limits, reading count, rule, evidence needed
INSPECTION_HEADER + READINGS	unit, inspector, check, sample, value, instrument, time
QUALITY_DISPOSITIONS	pass/fail/incomplete; release/scrap/hold; decision maker
DEFECT_RECORDS	defect, check, unit, detected by and at
CAUSE_INVESTIGATIONS	suspected and confirmed origin, cause, status, reviewer
RESPONSIBILITY_LINKS	party type and ID, relationship, evidence
BLOCKER_EVENTS	task, reason, duration, resolution

A4 · Synthetic data (3 of 3): control, outputs and record rules

E · Control and outputs

Table	Essential columns
ASSUMPTIONS_REGISTER	assumption, value, owner, status, source
SCENARIO_CATALOG	scenario ID, description, deterministic seed
EXPECTED_RESULTS	scenario, measure, value — calculated independently
ACTION_LOG	action, state (proposed, approved, started, completed, rejected, cancelled), who, when
FORECAST_RUNS	run ID, as-of time, inputs, outputs
Snapshots and alerts	derived worker, tank, station and order views; alerts; action comparisons

Rules for every record

- Stable IDs on every entity and event
- EVENT_TIME and KNOWN_AT on every event
- SCENARIO_ID, ORIGIN (synthetic/real), assumption references
- Units, record grain and meaning of empty values defined
- Ground truth (e.g. a future defect) kept apart from observable events

How the engine works

It advances through events — arrivals, work starts, completions, inspection decisions, transfers — applying opening WIP, qualified crew and minimum staffing, machines, parallel/batch capacity, setup and transport, breaks, inspection release, scrap, materials and replacements, queue limits, priorities and deadlines. Forecasts use only what is known at the as-of time.

A5 · Validation checks

Plain-language tests the demo must pass before anyone sees a number

- Scrapped tanks never proceed downstream
- Replacement tanks always get new IDs
- Material and unit quantities reconcile
- Workers are never double-booked
- Crew and equipment limits are respected
- Inspection outcomes agree with the readings
- Missing checks never become passes
- Defects and scrapped tanks are counted separately
- Replayed views never know the future
- Duplicate events never duplicate production
- Transfers always include the source-station effect
- Unfeasible recovery is shown honestly

A6 · Scenario catalogue with expected results

Scenario	Expected result (demo)
1 Normal shift	10 of 10 by 14:30; last 12:45
2 WC1 absence	10 of 10; last 13:55; Assembly waits 85 min
3 First-stage scrap	9 of 10 without replacement; never passes WC1
4 Combined disruption	9 of 10 by 14:30; Assembly waits 115 min
5 Protective WIP	3 opening tanks: Assembly's first wait at 08:30
6 Exhausted WIP	No approved tank at Assembly 09:55–11:35
7 Successful transfer	Arun + replacement: 10 of 10, last 14:21; no order late
8 Rejected transfer	Priya: Assembly below minimum crew → rejected

Scenario	Expected result (demo)
9 Feasible replacement	Scrap only: TANK-0021 done 13:25
10 Replacement material missing	Replacement can't start; stays 9 of 10; escalate
11 Machine breakdown	Parameters to agree; result to calculate
12 Delayed data	Fail 10:40, entered 10:48: replay at 10:45 shows no scrap
13 Confirmed worker cause	Person linked as cause only after confirmed review
14 Equipment cause	FX-02 confirmed; crew recorded "no fault"
15 Unresolved cause	Stays "unknown"; no person assigned

A7 · Founder Q&A (1 of 2)

What makes this different from a production dashboard?

A dashboard shows what is happening now. This calculates what happens next at each station and to the order, explains why, and tests recovery actions before they are approved.

Why do we need a production-flow engine?

Impact depends on queues, crew rules, breaks, inspection release and timing. Only stepping through events in order shows when a station runs out and when the order finishes.

Why can’t we copy the warehouse formulas?

Here a tank must pass gates in sequence, can be permanently scrapped, and needs routing, traceability and replacements. We reuse patterns, not formulas.

What can we prove with synthetic data?

That the rules are applied correctly, numbers reconcile and replay never knows the future. Not forecast accuracy in a real factory.

What requires real Siemens data?

Actual routing and gates, standard times, crew rules, WIP behaviour, disposition rules — and real tank histories to compare forecasts with outcomes.

How do we avoid blaming the wrong worker?

Detection, participation and cause are separate. A person is linked as cause only after a named reviewer confirms it with evidence. Unknown stays unknown.

A8 · Founder Q&A (2 of 2) and sources

What happens if no recovery action works?

We show it honestly: remaining shortfall, revised completion time, and which decision sits with the business (e.g. due date or overtime).

Does adding workers always help?

No — only when labour is the constraint. It doesn't fix missing material, a failed inspection or a broken machine, and it can slow the source station.

How will we measure pilot success?

With criteria agreed with Siemens up front — e.g. forecast timing versus actual events, alerts raised before impact, supervisor usability. No targets claimed yet.

What data should we request first?

WC1–WC4 routing and standard times, crew and qualification rules, scrap/disposition rules, and one real shift's tank event and inspection history.

How will this fit with existing factory systems?

Read-only from existing sources at first. Which systems hold this data is a question for Siemens; the application doesn't replace them.

Sources and limits of this deck

Used: our deck brief, our project-understanding note, two flow diagrams (tank route; cause investigation). Not available: CTQ workbook and the three repositories — related claims marked unverified. All numbers: our synthetic model.